

BRAINPOWER EDUCATION

Specialist Mathematics Units 1/2 - Matrices 2026

Assessment Guide / Marking Scheme - 40 marks

The printed Q4/Q5 headings disagree with their part-mark totals. This guide follows the printed part allocations; the full paper still totals 40 marks.

Q1a - 3

$A=B$ gives $x-y=2$, $x+y=10$, $w+1=4$ and $z^2=9$. Hence $x=6$, $y=4$, $w=3$, $z=\pm 3$.

Mark	Evidence required
1	Forms $x-y=2$ and uses $x+y=10$.
1	$x=6$, $y=4$.
1	$w=3$ and $z=\pm 3$.

Q1b - 1

$B^T = \begin{bmatrix} 2 & 0 \\ 4 & 9 \end{bmatrix}$.

Mark	Evidence required
1	Correct transpose.

Q1c - 1

$M+N=N+M$ for matrices of the same order.

Mark	Evidence required
1	States commutative law correctly.

Q2a - 3

$MP = \begin{bmatrix} 125 & 40 & 75 \end{bmatrix}^T$ million Wadiyan dollars.

Mark	Evidence required
1	Correct first row-by-column product.
1	Correct remaining products.
1	Final column vector $\begin{bmatrix} 125 & 40 & 75 \end{bmatrix}^T$.

Q2b - 2

P is 3×1 and M is 3×3 . PM is undefined because the inner dimensions 1 and 3 do not match.

Mark	Evidence required
1	States relevant orders/inner-dimension rule.
1	Concludes PM is undefined.

Q2c - 2

If A is $m \times n$ and B is $n \times p$, AB is $m \times p$.

Mark	Evidence required
1	Recognises AB is defined.
1	Order $m \times p$.

Q3a - 1

$[[1,2,1],[3,-1,2],[2,3,-1]][x,y,z]^T=[4,11,1]^T$.

Mark	Evidence required
1	Correct $AX=B$.

Q3b - 4

$\det(A)=-5+14+11=20$.

Mark	Evidence required
1	Valid cofactor expansion.
1	First minor contribution -5.
1	Remaining contributions 14 and 11.
1	$\det(A)=20$.

Q3c - 3

$x=2$, $y=-1/5$, $z=12/5$.

Mark	Evidence required
1	Valid exact solution method.
1	Obtains two correct exact values.
1	All three values correct.

Q4a - 3

$\det(S)=(m-2)(m-1)-6=(m-4)(m+1)$. Thus $m=4$ or -1 .

Mark	Evidence required
1	Correct determinant.
1	Correct factorisation.
1	$m=4, -1$.

Q4b - 3

For $m=5$, $S=[[3,2],[3,4]]$, $\det(S)=6$ and $S^{-1}=(1/6)[[4,-2],[-3,3]]$.

Mark	Evidence required
1	$\det(S)=6$.
1	Correct inverse structure.
1	Correct inverse.

Q5a - 2

Coefficient determinant $=(k-4)(k+1)$. Unique solution for k not equal to $4, -1$.

Mark	Evidence required
1	Finds determinant.
1	$k \neq 4, -1$.

Q5b - 4

$k=4$ gives $x+y=2$ and $x+y=1$: inconsistent. $k=-1$ gives $-4x+y=2$ and $4x-y=-1$: inconsistent. Hence no solution for $k=4$ or -1 .

Mark	Evidence required
1	Identifies $k=4, -1$.
1	Shows $k=4$ inconsistent.
1	Shows $k=-1$ inconsistent.
1	Correct conclusion.

Q5c - 3

No value of k gives infinitely many solutions; both singular cases are inconsistent.

Mark	Evidence required
1	States dependence/consistency requirement.
1	Uses both singular cases.
1	Concludes no such k .

Q6a - 3

One example is $B = \begin{bmatrix} -2 & 0 \\ 1 & 0 \end{bmatrix}$, for which $AB=0$ and B is non-zero.

Mark	Evidence required
1	Chooses columns in null space.
1	Valid non-zero B .
1	Verifies $AB=0$.

Q6b - 2

A and B are both non-zero but $AB=0$, so the Null Factor Law does not generally apply to matrices.

Mark	Evidence required
1	Notes non-zero factors with zero product.
1	Correct conclusion.